

SURAJ KUMAR SAHU

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EDUCATION

University of California Merced <i>PhD Candidate, Physics Advisor: Prof. Ajay Gopinathan</i>	Jan 2021 – Present <i>GPA: 3.84/4.00</i>
National Institute of Technology Rourkela <i>Master of Science in Physics</i>	Aug 2017 – May 2019
D.R. Nayapalli College, Utkal University <i>Bachelor of Science (Honors in Physics)</i>	Jul 2014 – Jun 2017

SKILLS

- **Computational Modeling & HPC:** Agent-Based Modeling, Numerical Simulations, Particle-Based Simulations, Reaction-Diffusion Systems, Network Dynamics, High-Performance Computing (HPC) with Slurm
- **AI & Agentic Research:** Agentic Research Workflows, AI-Assisted Computational Reproducibility, Context and Prompt Engineering, MCP Development (**Cellpose-MCP**), Custom Agents and Skills
- **Data Science & Visualization:** Data Analysis and Modeling, Scientific Visualization, Image Analysis, NumPy, SciPy, NetworkX
- **Programming Languages & Software:** Python, Julia, LaTeX, Git, Claude, Codex, VS Code, Cursor

PUBLICATIONS

Sahu, S., M. Biswas, “**Modeling protein association from homogeneous to mixed environments: A reaction-diffusion dynamics approach.**”, *Journal of Molecular Graphics and Modeling*, vol. 107, pp. 107936 (2021).

PROFESSIONAL EXPERIENCE

Graduate Research Assistant Gopinathan Group, UC Merced	<i>Jan 2021 – Present</i>
• Developed computational model of collagen matrix compaction in collaboration with Dasbiswas Lab. <i>In preparation.</i> • Developed an Active Gel Model of Cell Junction Mechanics: adhesion dynamics, contact formation, actin dynamics, cadherin pattern formation, and cell junction stability and rupture kinetics. <i>In preparation.</i> • Agent-based Model of Vascular Network Formation: cell migration, multicellular network formation, and remodeling; quantified network functionality, resilience, and adaptability. Collaboration with Sindi Lab & McCloskey Lab.	
Graduate Teaching Assistant Department of Physics, UC Merced	<i>Jan 2021 – Present</i>
• Led experimental labs and discussions for introductory physics courses; served as teaching assistant for upper-division course. • Graduate Teaching Assistant for PHYS 115, PHYS 009, PHYS 009L, PHYS 018, PHYS 008L across multiple terms (2021–2026).	
Graduate Student Researcher Computational Biophysics Group, NIT Rourkela	<i>Aug 2017 – May 2019</i>
• Coarse-grained reaction-diffusion (ReaDDy) modeling of macromolecular crowding effects on protein association; showed crowder size dominates over shape in stabilizing products; characterized additivity rules in mixed-crowder systems. Published in <i>J. Mol. Graph. Model.</i> (2021). • Developed nonlinear dynamical models of language competition among endangered scheduled Indian languages and estimated missing demographic data.	

GRADUATE COURSEWORK

- **Physics:** Classical Mechanics, Electrodynamics, Statistical Mechanics, Quantum Mechanics, Non-linear Dynamics and Chaos
- **Life Science:** Cell and Cellular Techniques, Basics in Molecular Medicine, Recombinant DNA Technology.
- **Computation:** Computational Physics, Classical Molecular Simulation, Numerical Mathematical Methods for Physics, Machine Learning & Statistics for Physics and Astronomy.

CONFERENCES & WORKSHOPS

Organizer & Instructor

-  **Graduate Division Workshop on AI Tools for Research and Data Analysis:** context engineering and agentic research workflows applied to scientific data analysis and image analysis pipelines; attended by graduate students.
- Led research project and instructed graduate trainees during the Center for Engineering and Mechanobiology annual bootcamp

Selected Oral Presentations:

- **BPS 2026**, San Francisco — “Collective Cell Motility of Fibroblasts Driven by Contractile Multicellular Network Formation”
- **APS March Meeting 2025**, Los Angeles — “Stability of Cell-Cell Junctions: Balancing Cortical Tension and Cadherin Aggregation”
- **Cell Bio 2024, ASCB/EMBO**, San Diego — “Balancing Cortical Tension and Adhesive Force for Stable Cell Junctions”
- **APS March Meeting 2024**, Minneapolis — “Modeling Cell-Cell Junction Mechanics in Vascular Networks”
- **APS March Meeting 2023**, Las Vegas — “Assembly and Mechanical Remodeling of Vascular Network”
- **APS March Meeting 2022**, Chicago — “Agent-Based Simulation of Vasculogenesis”

AWARDS & ACHIEVEMENTS

- **2025:** CEMB Summer Research Fellowship; CCBM Travel Award; Physics Graduate Group Travel Fellowship
- **2024:** GradExcel Peer Mentor Award; Physics Graduate Group Travel Fellowship
- **2023:** CCBM Outreach Fellowship; CCBM Travel Fellowship; Physics Graduate Group Travel Fellowship
- **2022:** Bobcat Summer STEM Academy Fellowship; Physics Graduate Group Travel Fellowship

LEADERSHIP, OUTREACH & COMMUNITY ENGAGEMENT

Graduate Student Research Mentor | I-BEST NSF REU, UC Merced

Jun – Aug 2026

- Mentored **Akshay Langhani**, a first-year Illinois State University undergraduate, on “Quantifying Biological Functionality of In Silico Vascular Networks.”

Trainee Leadership Council | Center for Engineering and Mechanobiology (CEMB)

Aug 2023 – Aug 2025

- Planned and organized tutorials, research presentations, and professional development workshops for CEMB graduate trainees across multiple institutions.

President | Graduate Biophysics Club, UC Merced

Jun 2021 – Jun 2023

- Led biophysics journal club, organized science outreach events and workshops, and coordinated professional development and networking opportunities for graduate students.

GradExcel Peer Mentor | UC Merced Graduate Division

Aug 2024 – Aug 2025

- Mentored incoming graduate students on personal well-being, academic success, and professional development; recognized with GradExcel Peer Mentor Award (2024).

Science Outreach Highlights:

- APS DSOFTE Grad School Panel Panelist (2025); Bahujan Scholars Network Panelist (Sep 2024); Foldscope workshop at Digital Nalanda, Wardha, India (Jan 2024); CEMB Science Outreach Presenter demonstrating mechanobiology tools for high school teachers (Aug 2023); Bobcat Summer STEM Academy Instructor (Jul 2022); The Franklin Institute mobile museum exhibit planning (Aug 2022, with Dr. Jayatri Das); CCBM Science Outreach Organizer (Jul 2023, Jun 2022) for CellPaint coronavirus outreach and Foldscope microorganism exploration; Mother/Daughter Science Camp Volunteer (Fall 2021, AAUW, UC Merced).

REFERENCES

Prof. Ajay Gopinathan 
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